

Government Intervention in Markets

**ECON 101: Introduction to
Microeconomics**

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Government Interventions in Markets

- Price Controls
 - Price Ceilings
 - Price Floors (Price Supports)
- Taxes
- Subsidies
- Import quotas and tariffs



Price Ceilings

A **price ceiling** or **price cap** is a regulation that makes it illegal to charge a price higher than a specified level.

When a price ceiling is applied to a housing market it is called a **rent ceiling**.

If the rent ceiling is set *above* the equilibrium rent, it has no effect. The market works as if there were no ceiling.

But a rent ceiling set *below* the equilibrium rent creates

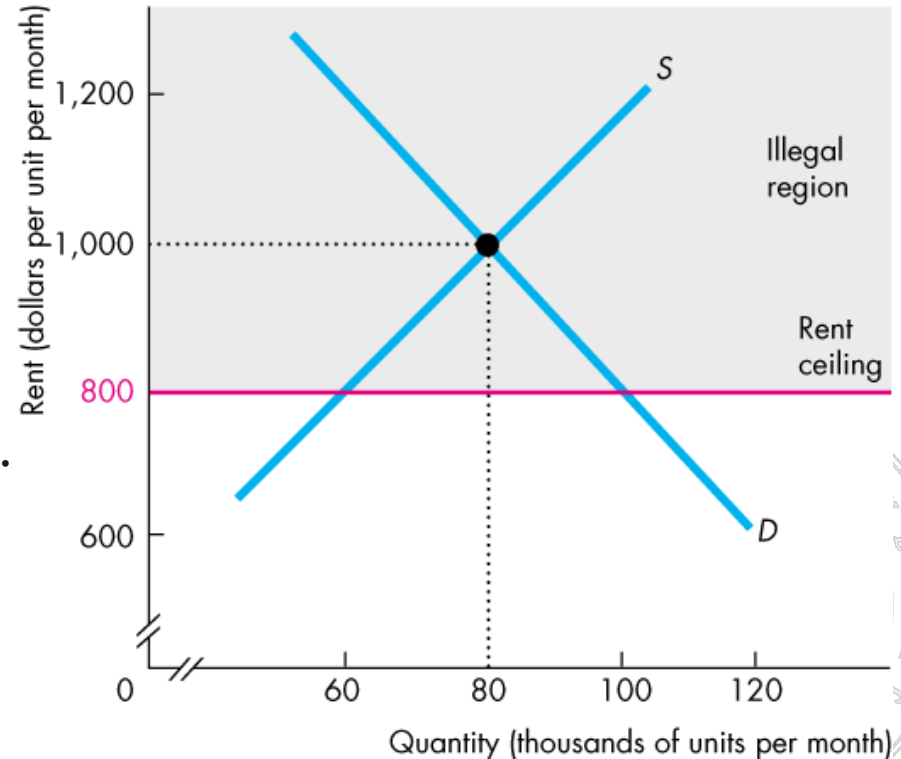
- A housing shortage
- Increased search activity
- A black market



A housing market with a Rent Ceiling

Housing Shortage

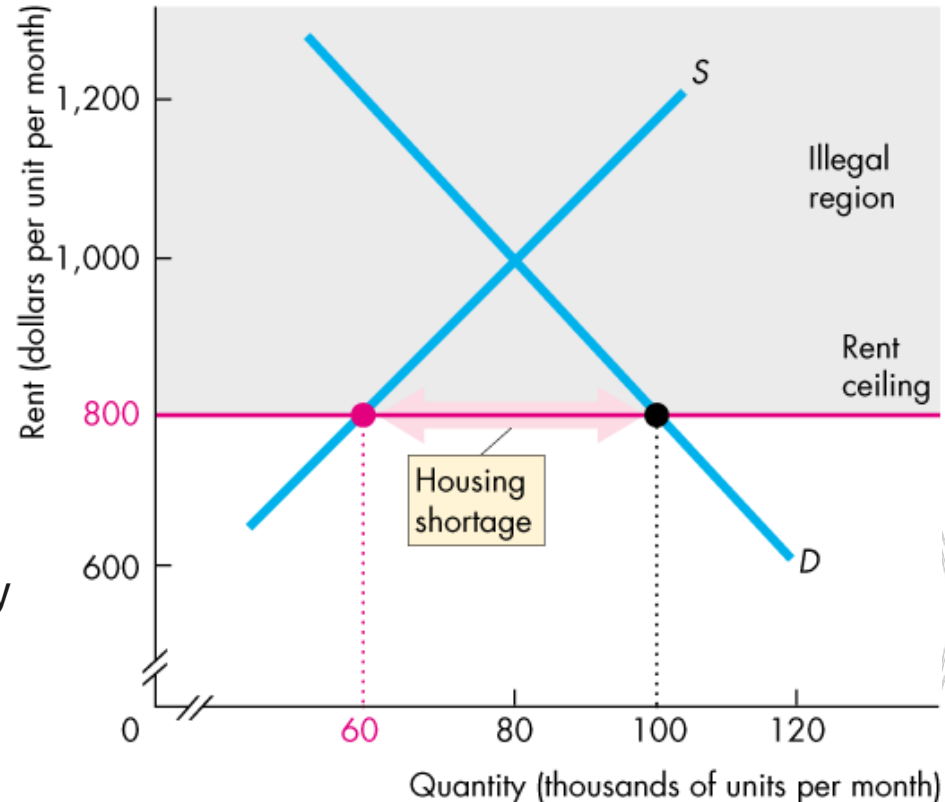
- The figure shows the effects of a rent ceiling that is set *below* the equilibrium rent.
- The equilibrium rent is \$1,000 a month.
- A rent ceiling is set at \$800 a month.
- So the equilibrium rent is in the illegal region.



A housing market with a Rent Ceiling

Housing Shortage

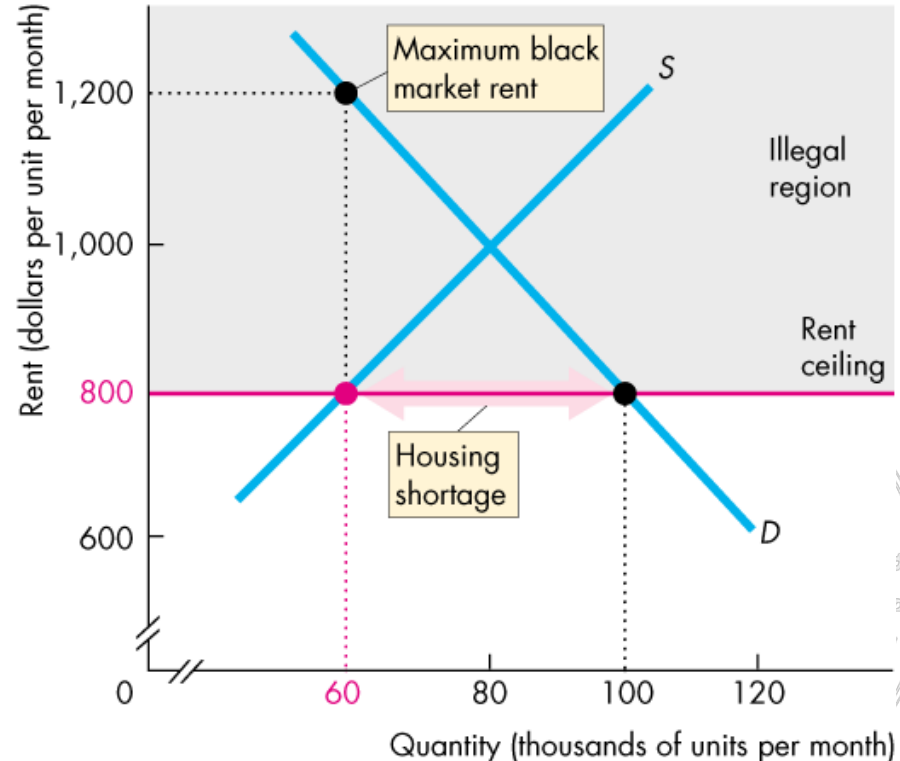
- The figure shows the effects of a rent ceiling that is set *below* the equilibrium rent.
- The equilibrium rent is \$1,000 a month.
- A rent ceiling is set at \$800 a month.
- So the equilibrium rent is in the illegal region.
- At the rent ceiling, the quantity of housing demanded exceeds the quantity supplied.
- There is a shortage of housing.



A housing market with a Rent Ceiling

Other mechanism operate to determine the price

- Because the legal price cannot eliminate the shortage, other mechanisms operate:
 - Increased search activity
 - A black market
- With the shortage someone is willing to pay up to \$1200 a month.



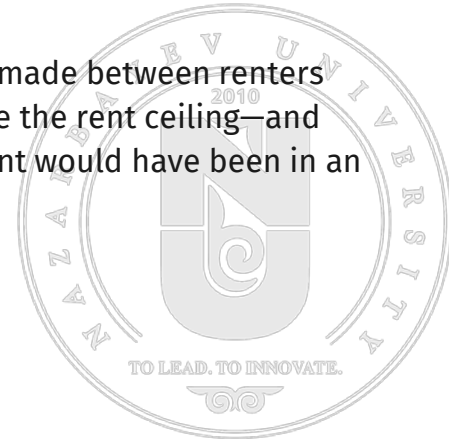
A Housing Market with a rent ceiling

Increased Search Activity

- The time spent looking for someone with whom to do business is called **search activity**.
- When a price is regulated and there is a shortage, search activity increases.
- Search activity is costly, and the opportunity cost of housing equals its rent (regulated) plus the opportunity cost of the search activity (unregulated).
- Because the quantity of housing is less than the quantity in an unregulated market, the opportunity cost of housing exceeds the unregulated rent.

A black market

- A **black market** is an illegal market that operates alongside a legal market in which a price ceiling or other restriction has been imposed.
- A shortage of housing creates a black market in housing.
- Illegal arrangements are made between renters and landlords at rents above the rent ceiling—and generally above what the rent would have been in an unregulated market.

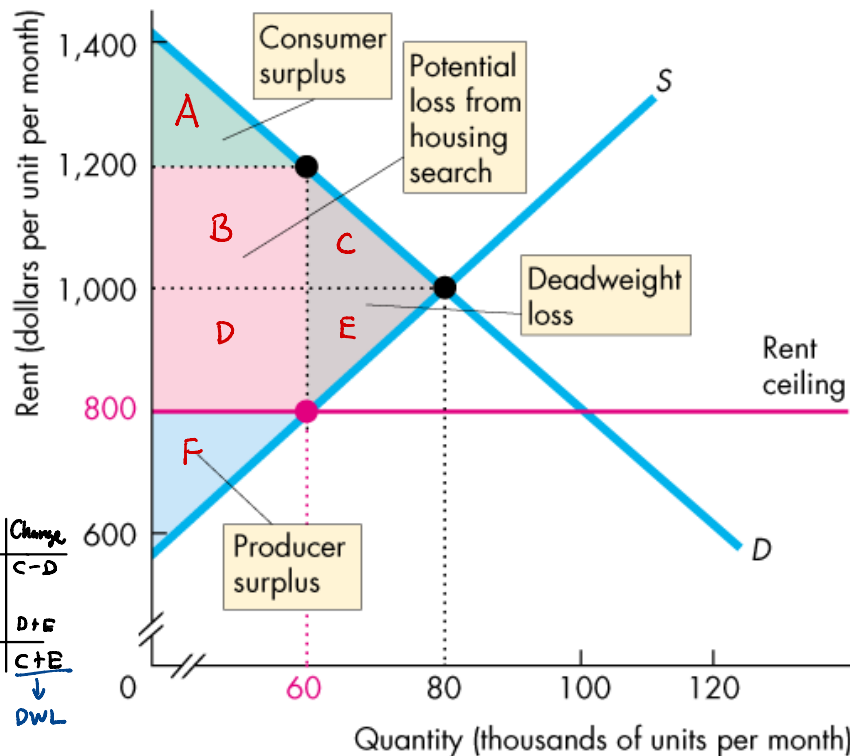


A Housing Market with a rent ceiling

Inefficiency of a Rent Ceiling

- A rent ceiling set below the equilibrium rent leads to an inefficient underproduction of housing services.
- The marginal social benefit from housing services exceeds its marginal social cost and a deadweight loss arises.
- A rent ceiling decreases the quantity of housing supplied to less than the efficient quantity.
- A deadweight loss arises.
- Producer surplus shrinks.
- Consumer surplus shrinks.
- There is a potential loss from increased search activity.

	No Ceiling	Ceiling	Change
CS	A+B+C	A+B+D	C-D
PS	D+E+F	F	D+E
SS	A+B+C+D+E+F	A+B+D	C+E ↓ DWL



TO LEAD. TO INNOVATE.

A Housing Market with a rent ceiling

Are Rent ceilings fair?

- From the point of view of *fair rules of exchange*, a rent ceiling is unfair because it blocks voluntary exchange.
- In terms of *results*, a rent ceiling is unfair because it does not generally benefit the poorest potential renters, i.e., the ones with the lowest willingness to pay because of their income.
- A rent ceiling decreases the quantity of housing and the scarce housing is allocated by
 - Lottery (random assignment)
 - First-come, first-served
 - Discrimination
- None of this methods leads to a fair outcome!



Price floors

A **price floor (or price support)** is a regulation that makes it illegal to trade at a price lower than a specified level.

When a price floor is applied to labor markets, it is called a **minimum wage**.

If the minimum wage is set *below* the equilibrium wage rate, it has no effect. The market works as if there were no minimum wage.

If the minimum wage is set *above* the equilibrium wage rate, it has powerful effects.



A Labor Market with a Minimum Wage

Minimum Wage Brings Unemployment

If the minimum wage is set *above* the equilibrium wage rate, the quantity of labor supplied by workers *exceeds* the quantity demanded by employers.

There is a surplus of labor.

The quantity of labor hired at the minimum wage is *less* than the quantity that would be hired in an unregulated labor market.

Because the legal wage rate cannot eliminate the surplus, the minimum wage creates unemployment.



A Labor Market with a Minimum Wage

The equilibrium wage rate is \$6 an hour.

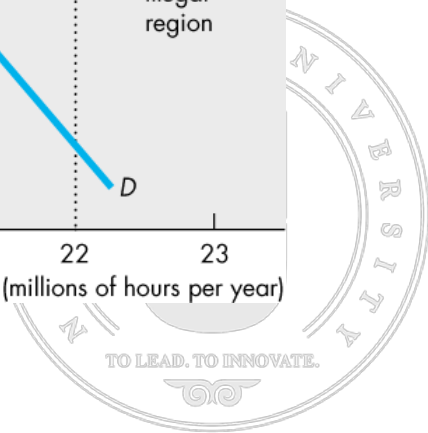
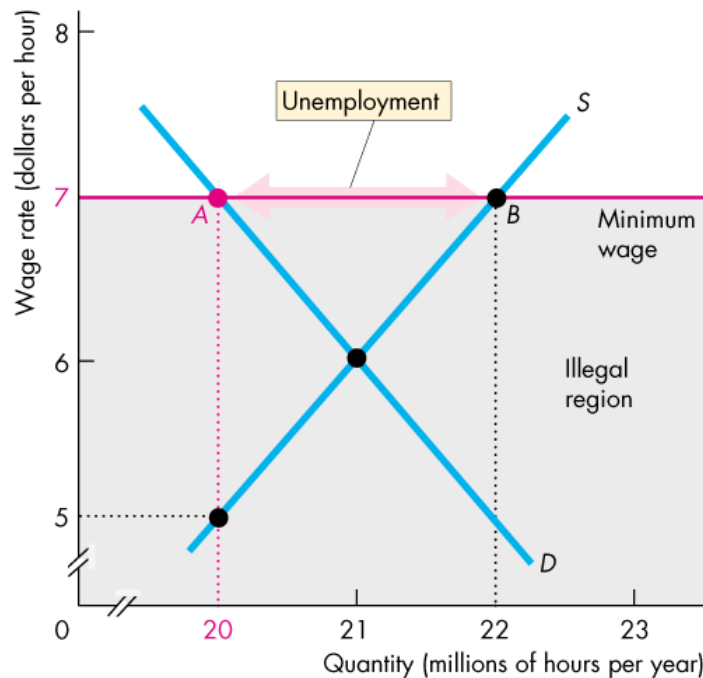
The minimum wage rate is set at \$7 an hour.

So the equilibrium wage rate is in the illegal region.

The quantity of labor employed is the quantity demanded.

The quantity of labor supplied exceeds the quantity demanded and unemployment is created.

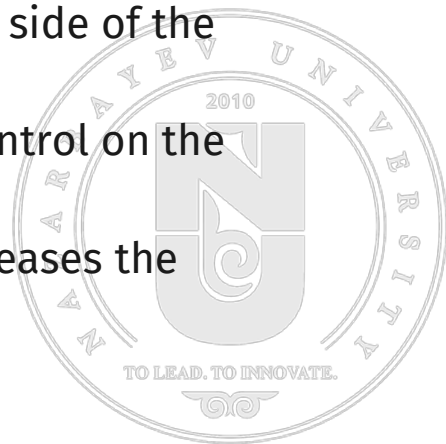
With only 20 million hours demanded, some workers are willing to supply the last hour demanded for \$5.



A Labor Market with a Minimum Wage

Is a minimum wage fair?

- This is a contested question with not an easy answer.
- On the one hand, it is a fact that the minimum wage restricts employment by introducing an artificially high wage level.
- Whether it creates unemployment, depends on the structure of the market in which the minimum wage is applied: the bargaining power of each side of the market pre-minimum wage is crucial.
- However, it is generally agreed that *ceteris paribus*, an artificial control on the wage will create unemployment.
- Specifically, most economists believe that the minimum wage increases the unemployment rate of low-skilled younger workers.

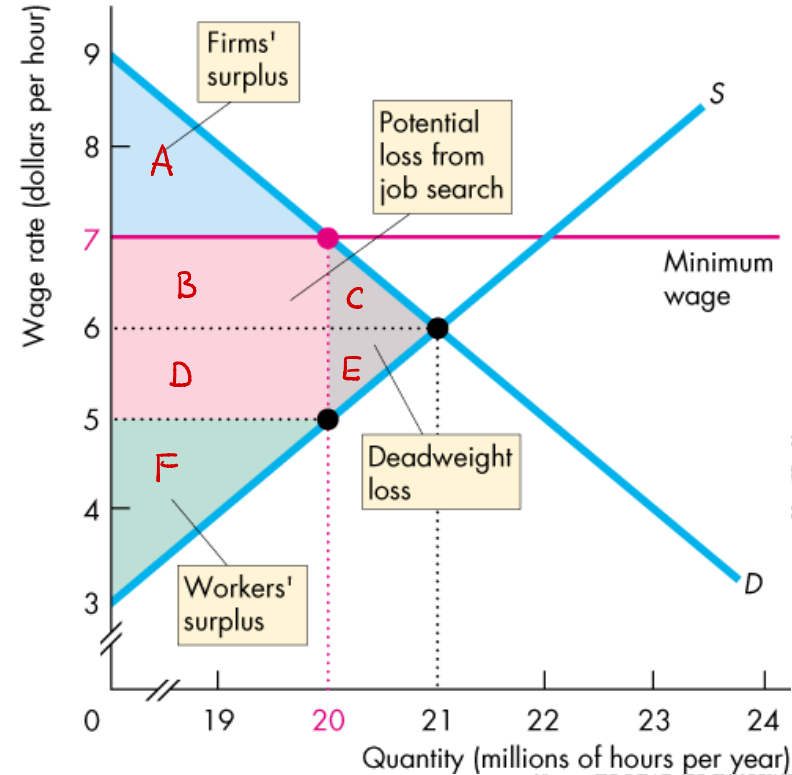


A Labor Market with a Minimum Wage

	No min wage	Min wage	Δ
CS	$A+B+C$	A	$B+C$
PS	$D+E+F$	$B+D+F$	$E-B$
SS	$A+B+C+D+E+F$	$A+B+D+F$	$C+E$

Inefficiency of a Minimum Wage

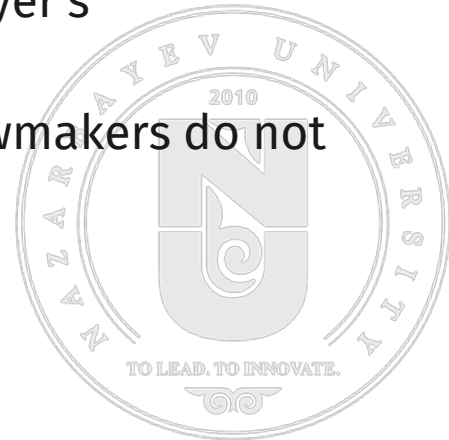
- A minimum wage leads to an inefficient outcome.
- The quantity of labor employed is less than the efficient quantity.
- The supply of labor measures the marginal social cost of labor to workers (leisure forgone).
- The demand for labor measures the marginal social benefit from labor (value of goods produced).
- A minimum wage set *above* the equilibrium wage decreases the quantity of labor employed.
- A deadweight loss arises.
- The potential loss from increased job search decreases both workers' surplus and firms' surplus.



TO LEAD. TO INNOVATE.

Taxes

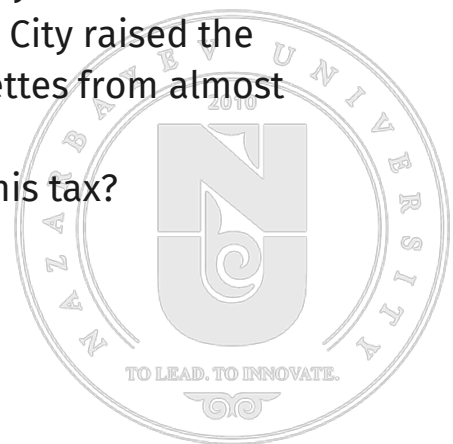
- Everything you earn and most things you buy are taxed.
- Who really pays these taxes?
- Income tax and the Social Security tax are deducted from your pay, and sales tax is added to the price of the things you buy, so isn't it obvious that you pay these taxes?
- Isn't it equally obvious that your employer pays the employer's contribution to the Social Security tax?
- It turns out it is not obvious who **actually pays** and that lawmakers do not decide who ends up paying.
- **Economic vs Legal Incidence.**
- Two types of taxes: per-unit and ad-valorem taxes.



Tax Incidence

Tax Incidence

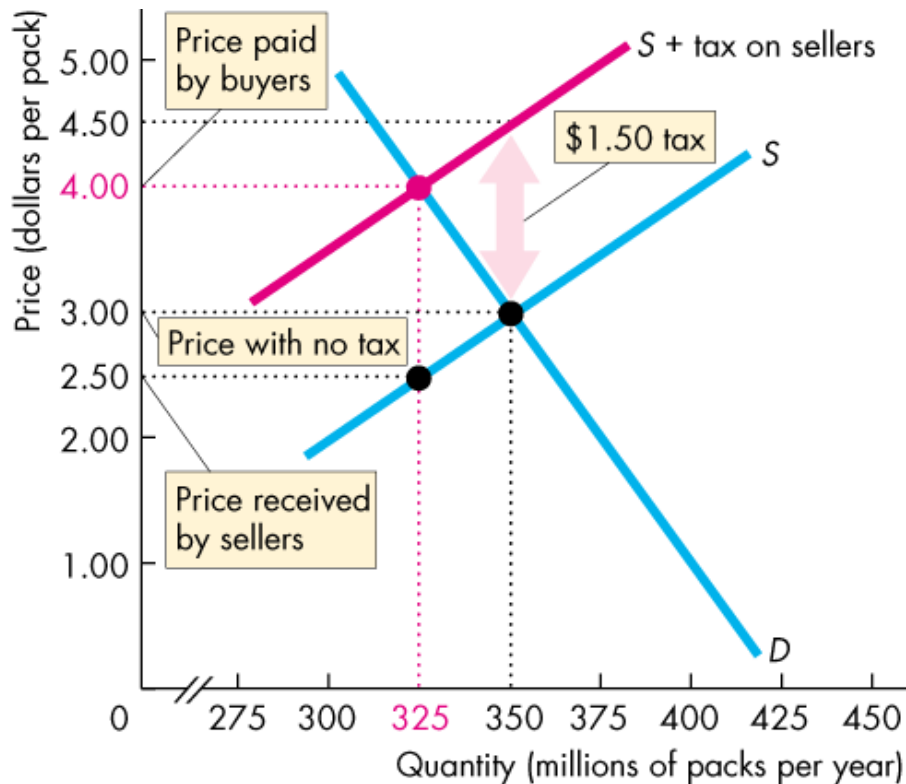
- Tax incidence is the division of the burden of a tax between buyers and sellers.
- When an item is taxed, its price might rise by the full amount of the tax, by a lesser amount, or not at all.
- If the price rises by the full amount of the tax, buyers pay the tax.
- If the price rises by a lesser amount than the tax, buyers and sellers share the burden of the tax.
- If the price doesn't rise at all, sellers pay the tax.
- Tax incidence doesn't depend on tax law!
- The law might impose a tax on buyers or sellers, but the outcome will be the same.
- To see why, we look at the tax on cigarettes in New York City.
- On July 1, 2002, New York City raised the tax on the sales of cigarettes from almost nothing to \$1.50 a pack.
- What are the effects of this tax?



Taxes

A Tax on Sellers

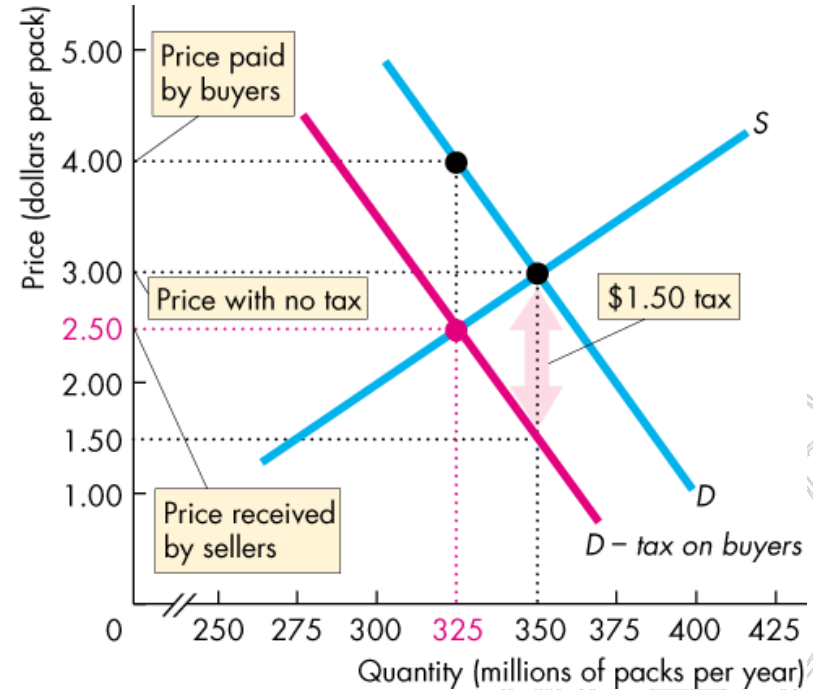
- The Figure shows the effects of this tax.
- With no tax, the equilibrium price is \$3.00 a pack.
- A tax on sellers of \$1.50 a pack is introduced.
- Supply decreases and the curve $S + \text{tax on sellers}$ shows the new supply curve.
- The market price paid by buyers rises to \$4.00 a pack and the quantity bought decreases.
- The price received by the sellers falls to \$2.50 a pack.
- So with the tax of \$1.50 a pack, buyers pay \$1.00 a pack more and sellers receive 50¢ a pack less.



Taxes

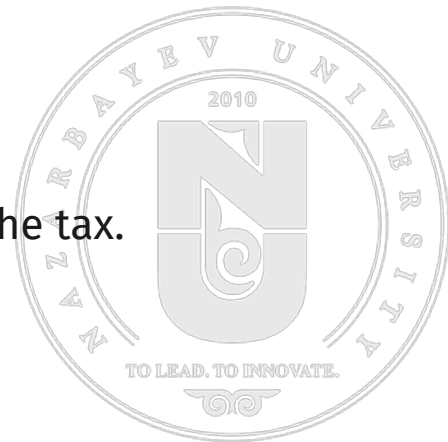
A Tax on Buyers

- Again, with no tax, the equilibrium price is \$3.00 a pack.
- A tax on buyers of \$1.50 a pack is introduced.
- Demand decreases and the curve $D - \text{tax on buyers}$ shows the new demand curve.
- The price received by sellers falls to \$2.50 a pack and the quantity decreases.
- The price paid by buyers rises to \$4.00 a pack.
- So with the tax of \$1.50 a pack, buyers pay \$1.00 a pack more and sellers receive 50¢ a pack less.



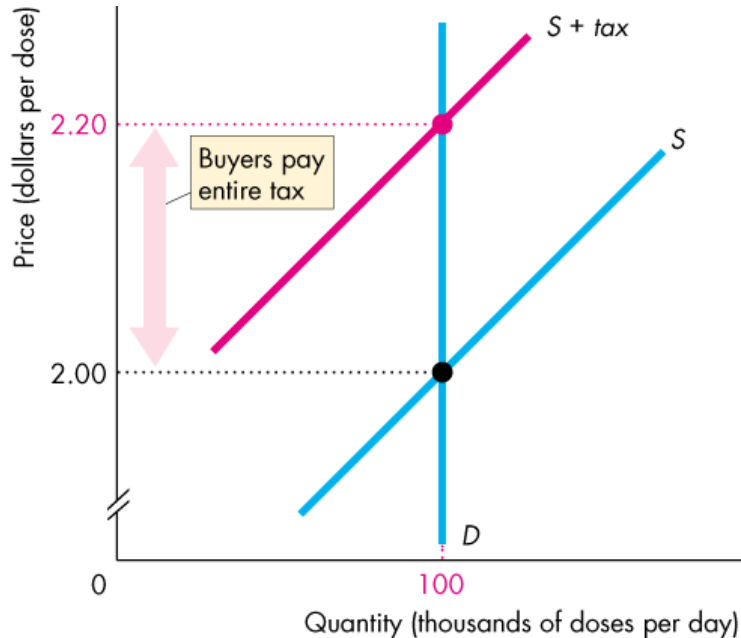
Taxes

- The lesson to learn from the previous example is that the **Economic tax incidence remains the same, irrespective of whether the legal incidence (law) says sellers pay or buyers pay.**
- The division of the tax between buyers and sellers depends on the elasticities of demand and supply.
- To see how, we look at two extreme cases.
 - Perfectly inelastic demand: Buyers pay the entire tax.
 - Perfectly elastic demand: Sellers pay the entire tax.
- The more inelastic the demand, the larger is the buyers' share of the tax.

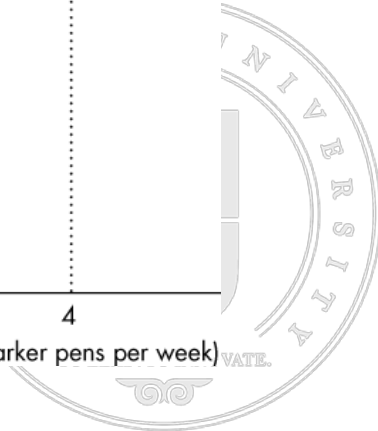
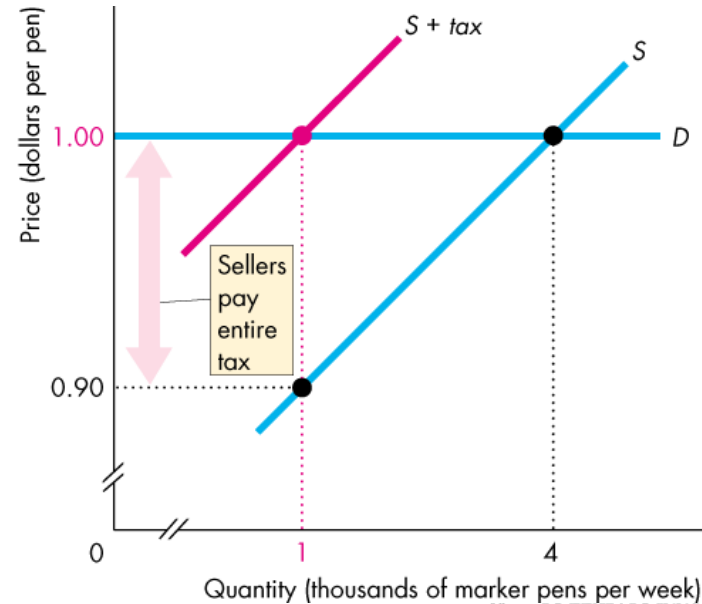


Taxes

Perfectly Inelastic Demand



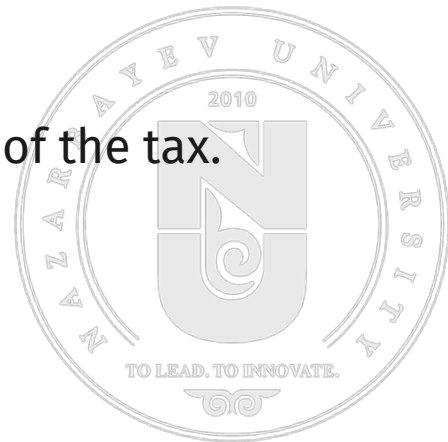
Perfectly Elastic Demand



Taxes

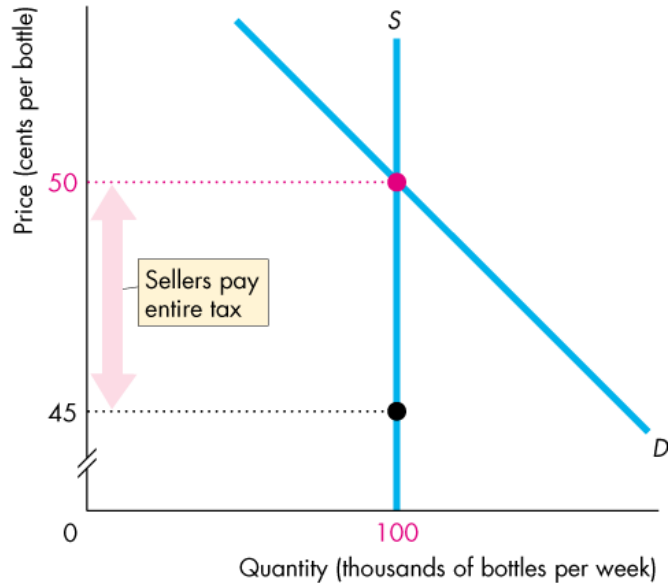
Tax Incidence and Elasticity of Supply

- To see the effect of the elasticity of supply on the division of the tax payment, we again look at two extreme cases.
 - Perfectly inelastic supply: Sellers pay the entire tax.
 - Perfectly elastic supply: Buyers pay the entire tax.
- The more elastic the supply, the larger is the buyers' share of the tax.



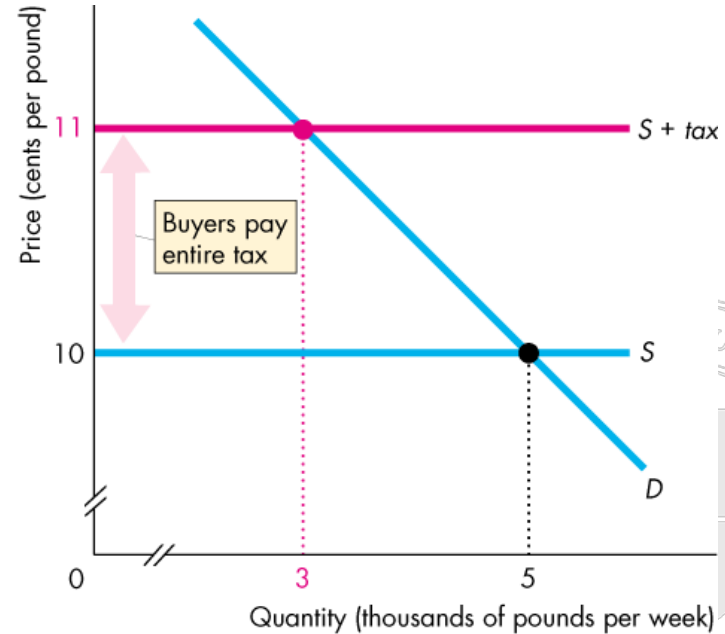
Taxes

Perfectly Inelastic Supply



(a) Perfectly inelastic supply

Perfectly Elastic Supply



(b) Perfectly elastic supply

Taxes

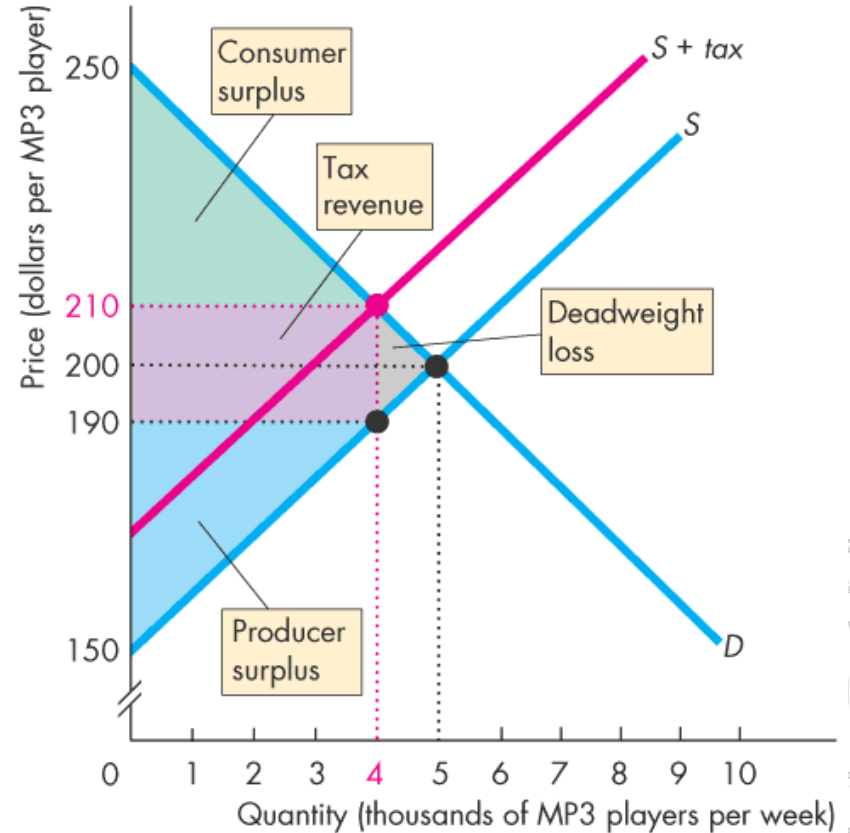
Taxes and Efficiency

- Whenever a tax restricts the quantity that is traded with respect to the unrestricted equilibrium price, an inefficiency will arise.
- Except in the extreme cases of perfectly inelastic demand or perfectly inelastic supply when the quantity remains the same, imposing a tax creates inefficiency.



Taxes

- With no tax, marginal social benefit equals marginal social cost.
- Total surplus (the sum of consumer surplus and producer surplus) is maximized.
- The market is efficient.
- The tax decreases the quantity, raises the buyers' price, and lowers the sellers' price.
- Marginal social benefit exceeds marginal social cost and the tax is inefficient.
- The tax revenue takes part of the total surplus.
- The decreased quantity creates a deadweight loss.



Subsidies and Production Quotas

- A **production quota** is an upper limit to the quantity of a good that may be produced during a specified period.
- A **subsidy** is a payment made by the government to a producer.



Subsidies

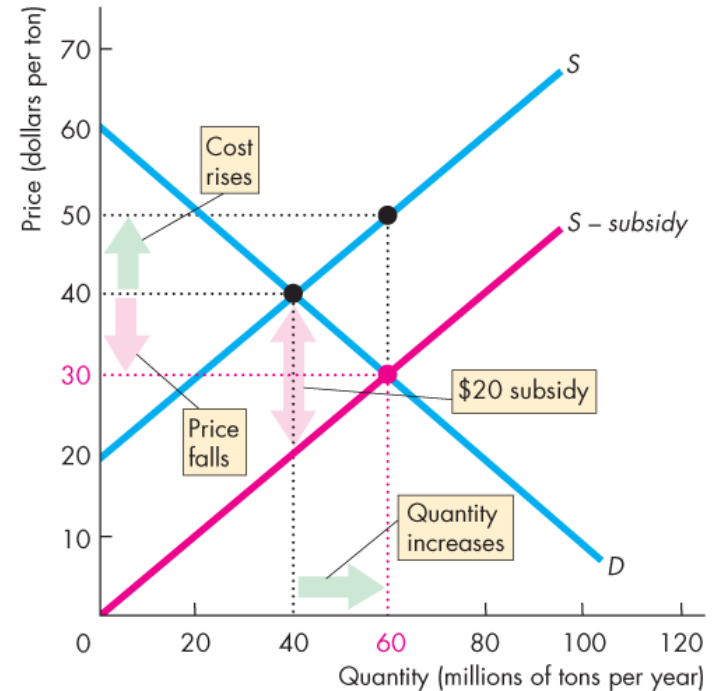
A subsidy of farm products

- With no subsidy, the price is \$40 a ton and 40 million tons a year are produced.
- With a subsidy of \$20 a ton, marginal cost minus subsidy falls by \$20 a ton and the new supply curve is $S - \text{subsidy}$.
- The market price falls to \$30 a ton and farmers increase the quantity to 60 million tons a year.
- But farmers' marginal cost increases to \$50 a ton.
- With the subsidy, farmers receive more on each ton sold—the price of \$30 a ton plus the subsidy of \$20 a ton, which is \$50 a ton.

Inefficient Overproduction

At the quantity produced:

- Consumers' willingness to pay equals the market price, which has fallen.
- Producers' willingness to sell has increased and exceeds willingness to pay from consumers.





You can read about this in chapters 2 and 9 from Perloff.